

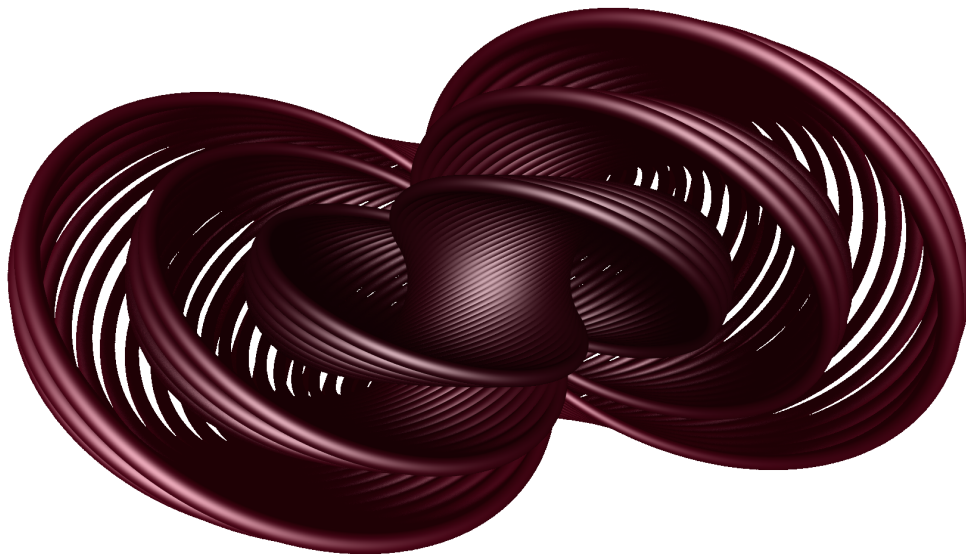
The Hopf Fibration and the Qubit

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ABSTRACT

In the present work, we introduce the basic notions of the qubit and the quantum systems that represent it, i.e., the two-level quantum systems. This is done in two ways: solving the eigenvalue problem and via the Bloch representation. Also, we briefly introduce the notion of a quantum gate.

Furthermore, we develop a series of maps, following the constraints advised by quantum mechanics, to find a relation between the spheres of three and two dimensions. The map obtained is the Hopf map, which defines a principal and a fiber bundle. Therefore, we introduce some of the theory behind these concepts and prove that the Hopf map is indeed a principal and a fiber bundle with group $U(1) \cong \mathbb{S}^1$ over the total space $\mathbb{S}^3$ with base space $\mathbb{S}^2$ —the so-called Hopf fibration. We conclude with some visualizations of the fibers or quantum states associated with the map.

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INTRODUCTION

The fundamental unit in information science is the bit, representing a digit in the binary numeral system. A bit can take on two possible values, "0" and "1". The quantum analog of a bit is the qubit (quantum bit), a two-level quantum system[1]. In fact, almost any two-state quantum system can be used to represent a qubit[2, 3]. For instance, some of these two-level quantum systems have been experimentally realized in quantum optics where various manipulation protocols already exist. The study and experimental manipulation of qubits is a rapidly developing area within quantum information physics. Generally, a quantum system with n levels can be studied as qudits. One of the most important properties of quantum systems that model a qubit is the ability to exhibit coherent state superposition[4, 5]. Thus, the principle of superposition plays a significant role in the study of quantum information [6].

A fundamental aspect in the study of qubits, at least for the purposes of this work, is their relationship with geometry and topology. For a two-level quantum system, it is well known in quantum optics that the Bloch sphere representation is used to describe the state of a qubit, up to a global phase, by a point on a 2-sphere[7]. In this representation, the coordinates are the expectation values of the physical operators of interest for a given quantum state. Another fundamental fact is the relationship between the Bloch sphere and the Hopf fibration of the 3-sphere[8, 9, 10, 11]. The Hopf fibration is defined via a mapping from the 3-sphere to the 2-sphere, known as the Hopf map. Historically, it is the first map from a higher-dimensional sphere to a lower-dimensional one that is not homotopically trivial, demonstrating that the third homotopy group of the 3-sphere is nontrivial[12]. This fact pushes forward the development of homotopy theory and the calculation of higher-order homotopy groups. In particular, these ideas lead to the demonstration that, in general, the k -th homotopy group of the n -sphere can be nontrivial for k greater than n , which contrasts with homology groups where it is known that the k -th homology group of an n -sphere is trivial for k greater than n [13, 14].

Associated with the Hopf map, it is possible to assign a homotopy invariant. There are various ways to define it; however, according to a fundamental theorem by F. Adams and later by M. Atiyah using K-theory methods, the Hopf invariant for the Hopf map between the 3-sphere and the 2-sphere is 1. From a geometric perspective, the fibers of the Hopf fibration are circles. There exists an infinity of circles in the 3-sphere, and any two fibers are linked with a linking number equal to one, which is, in fact, the Hopf invariant [15]. The Hopf fibration provides a homotopic way to visualize the 3-sphere in terms of disjoint circles. Given this context, it is important to explicitly highlight the significance of the Hopf fibration in Lie group theory.

As previously mentioned, the Hopf fibration is an important object in mathematics, but it also plays a prominent role in physics. It has numerous applications, including quantum information with the study of n -state quantum systems, classical mechanics with the study of harmonic oscillators, general relativity with the study of spatially homogeneous cosmological models, differential and twistor geometry with Robinson congruences, and another application arises from the relationship between complex line bundles associated with the Hopf fibration and the zero-mass helicity representations of the Poincaré group. In field theory, it appears in the study of magnetic monopoles and the Dirac equation [16, 17]. For further reference on these applications, see [18].

The objective of this thesis is to provide a detailed review of a two-level quantum system, as well as its relationship with the Bloch sphere and the Hopf fibration.

This thesis is organized as follows. Chapter one provides a description of the concept of a qubit, introduces the Pauli matrices, and then reviews the two-level quantum system and the Bloch sphere representation. Finally, the concept of a quantum gate is briefly presented.

In Chapter 2, the main elements of the Hopf representation of the qubit are presented. Key geometric and topological concepts are introduced and finally applied to the description of the two-level quantum system.

Finally, Chapter 3 presents the conclusions obtained in this work.

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Quiero utilizar este pequeño espacio para agradecer el apoyo de aquellas personas que siempre me han ofrecido su ayuda incondicional. Ese es el caso de mis padres, los cuales me han dado su apoyo en cualquier sueño que les presente y que sus enseñanzas me han formado en la persona que soy ahora. A mis amigos Ramsses y Benjamín, los cuales a lo largo de la carrera me ofrecieron su amistad y me ayudaron a continuar hasta el final. A los profesores Germán Santos, Ignacio Otero y Myriam Maldonado, los cuales me tendieron una mano cuando los necesite y me enseñaron el hermoso camino de las matemáticas. Por último, pero no menos importante, a mis asesores, el Prof. Pablo Paniagua, el Prof. Francisco Turrubiates y al Prof. Marcelo Aguilar, los cuales me permitieron llevar a cabo de manera satisfactoria el presente trabajo.

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1 | THE PHYSICAL REPRESENTATION OF THE QUBIT

For many years, information theory, the branch that studies quantification, storage, and communication of information, had remained classical. This last means that many of the concepts of communication and computer science remained under the ideas of classical mechanics. For example, the Turing machine is a classical mechanical model that behaves according to purely classical mechanics principles. Even when classical physics was left behind in modern computational devices and hardware based on quantum mechanical phenomena arose. Information theory continued to be more than enough, and it accomplished this via abstractions that let us study many foundational ideas without worrying about the physical devices that make their connection to the physical world. Therefore, physics was kept to low-level advances. This contrast between physical and theoretical ideas was thought-provoking; quantum mechanics was the physical theory that led us to the modern-day devices we have now. But, the theory that used them was not.

“Quantum mechanics underlies the working of traditional, classical computer and communication devices, from the transistor through the laser to the latest hardware advances that increase the speed and power and decrease the size of computer and communication components” [3].

However, in the 1980s, researchers like Charles Bennett and Gilles Brassard started visualizing specific applications to classical information theory. For example, they showed that some peculiar properties of quantum systems provided a new and more secure way of encryption via a novel mechanism of establishing a cryptographic key. Another example lies in the hands of the famous physicist Richard Feynman. In 1981, he noted that one key feature of quantum mechanics, quantum entanglement, could not be simulated by a classical Turing machine and proposed doing it via a novel device, a quantum computer. These and other realizations of the applicability of quantum mechanical ideas to the way we think about information processing opened the door to what we will call ***quantum information theory***.

In general, quantum information processing differentiates from its classical counterpart by using quantum mechanics instead of classical mechanics to model information and its processing. Note that quantum computing is not about changing the physical principles used in computers but rather changing the notion of computing itself. Whether the computer is quantum or classical depends on whether the information is represented and manipulated in a quantum or classical way.

The fundamental transformation begins by realizing a new fundamental computation unit, the quantum bit or qubit, instead of the classical bit. In this chapter, we will review its physical construction. For an introductory reference of the development of the qubit, you can look at [1, 2, 3, 4, 5].

1.1 The qubit

Quantum bits, as we mentioned, are the fundamental units of information in the context of quantum information, and they take the same role as the bit in classical information processing. Keep in mind that in the same way in which the bit can be represented by voltage levels, magnetic states, optical states, or physical switches, the qubit can be “crafted” physically via many representations.

As in the case of classical information theory, it is advantageous for the theoretical treatment to not pay too much attention (at least at the fundamental level) to the physical realizations of the theory. The advantage of treating the qubit as an abstract construction is that we can build a quantum information theory without constraining our ideas to a particular physical representation [1].

Therefore, in general, the qubit has two basic possible states $|0\rangle, |1\rangle$; those states are the quantum analogs of 0, 1 of the classical bit. The key difference between the classical bit and the quantum bit is that in the bit, the only states available are 0, 1. Meanwhile, the quantum qubit can access any linear combinations of the original states $|0\rangle, |1\rangle$, i.e.,

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad a, b \in \mathbb{C}. \quad (1.1)$$

In other words, the qubit has infinite possible states, a continuum of states between $|0\rangle$ and $|1\rangle$, i.e., all the possible superpositions of those states- until measure. However, the quantum mechanical nature of the qubit restricts how we can examine it to determine if it's in the state $|0\rangle$ or $|1\rangle$. In other words, it tells us that when we measure the qubit, we will have $|0\rangle$ with probability $\|a\|^2$, and $|1\rangle$ with probability of $\|b\|^2$, where the following condition must be satisfied: $\|a\|^2 + \|b\|^2 = 1$. Furthermore, eq. (1.1) tells us a thought-provoking feature of the qubit: Any two-dimensional complex vector space that describes a quantum system can represent a qubit. Normally, it's labeled two-state systems, which we will study briefly in the following sections. From a practical perspective, it is yet unclear which two-state systems will be most suitable for physical realizations of quantum information processing devices such as quantum computers; a variety of physical representations of qubits will likely be used [3].

1.2 Pauli Matrices

To develop some of the theory behind the qubit, let us introduce a set of quantities that are pretty useful in the analysis. The Pauli matrices are a set of three essential matrices that appear naturally in the study of a particular two-state system named spin 1/2. These matrices

are essential because the operators associated with physical variables in the case of a two-level system can be written in terms of these matrices and the identity matrix. We will begin introducing some of their properties. Therefore, the Pauli matrices are defined by

$$\hat{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

which can be easily shown to be Hermitian matrices, i.e., that they are equal to their complex conjugate transpose:

$$\hat{\sigma}_1^\dagger = \hat{\sigma}_1; \quad \hat{\sigma}_2^\dagger = \hat{\sigma}_2; \quad \hat{\sigma}_3^\dagger = \hat{\sigma}_3.$$

And they obey the following commutation relations.

$$[\hat{\sigma}_i, \hat{\sigma}_j] = 2i\varepsilon_{ijk}\hat{\sigma}_k.$$

They are also traceless,

$$\text{Tr}\hat{\sigma}_i = 0.$$

Furthermore, the Pauli matrices have the property of being their own inverse, i.e.,

$$\hat{\sigma}_i^2 = I.$$

The most important feature is that, including the identity, let's call it the zeroth Pauli matrix σ_0 , they form a basis for the vector space of operators 2×2 . Let us prove that. For instance, note that the following equation holds for any pair of Pauli matrices and the identity,

$$\text{Tr}(\hat{\sigma}_\alpha \hat{\sigma}_\beta) = 2\delta_{\alpha\beta}.$$

Remember that the usual inner product associated with the vector spaces of matrices is the Frobenius inner product given by [19],

$$\langle A | B \rangle = \text{Tr}(AB^\dagger).$$

So that if

$$0 = \sum_{\alpha} c_{\alpha} \hat{\sigma}_{\alpha},$$

the coefficient c_{β} is given by

$$0 = \langle \hat{\sigma}_{\beta} | \sum_{\alpha} c_{\alpha} \hat{\sigma}_{\alpha} \rangle = \sum_{\alpha} \bar{c}_{\alpha} \text{Tr}(\hat{\sigma}_{\beta} \hat{\sigma}_{\alpha}) = \sum_{\alpha} 2\bar{c}_{\alpha} \delta_{\alpha\beta} = 2\bar{c}_{\beta}.$$

Therefore, $c_{\beta} = 0$ for any β . And since $\mathcal{M}_{2 \times 2}(\mathbb{C})$ is a 4-dimensional vector space, the set $\{\hat{\sigma}_0, \hat{\sigma}_1, \hat{\sigma}_2, \hat{\sigma}_3\}$ is a basis.

1.3 Introduction to 2-level quantum systems

Now, we can introduce and solve the quantum system we care about. A two-level system consists of two energy levels or two states, such as the electron spin and the qubit. These kinds of systems have the feature of being two-dimensional Hilbert spaces. We are going to follow the general procedure to solve this kind of quantum system [20, see pp. 223-229].

A two-state quantum system has to be given by a Hamiltonian operator given by a 2×2 Hermitian matrix (recall that the algebraic multiplicity of the eigenvalues constrains the geometric multiplicity). Then, the Hamiltonian of the system is written by

$$\hat{H} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix},$$

which satisfies $\hat{H} = \hat{H}^\dagger$: meaning that $H_{11}, H_{22} \in \mathbb{R}$, $H_{21}^* = H_{12} = H_3$. To find the general eigenvalues of the system, we need to solve the following equation:

$$\det(\hat{H} - \lambda \hat{I}) = \det\left(\begin{pmatrix} H_1 & H_3 \\ H_3^* & H_2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right) = 0,$$

that means solving the second-order polynomial given by

$$\lambda^2 - (H_1 + H_2)\lambda + H_1H_2 - \|H_3\|^2 = 0,$$

which can be solved directly using the general formulae of roots of second-order polynomials, giving

$$\lambda_{0,1} = \frac{1}{2} \left[(H_1 + H_2) \pm \sqrt{(H_1 - H_2)^2 + 4\|H_3\|^2} \right].$$

As expected, we have two eigenvalues, each with an associated eigenstate. We will use the customary notation E_0, E_1 for the eigenvalues and $|0\rangle, |1\rangle$ for the associated eigenstates. With this notation, the eigenvalue equation of the Hamiltonian is written

$$\hat{H}|0, 1\rangle = E_{0,1}|0, 1\rangle,$$

where $|0, 1\rangle$ stands for both possibilities, $|0\rangle$ or $|1\rangle$.

In the previous section, we introduced a basis for any 2×2 Hermitian operator whose elements were the identity matrix and the Pauli matrices. We can rewrite our Hamiltonian operator as a linear combination of these matrices in the following form

$$\hat{H} = d_0\hat{\sigma}_0 + d_1\hat{\sigma}_1 + d_2\hat{\sigma}_2 + d_3\hat{\sigma}_3,$$

with $d_0, d_1, d_2, d_3 \in \mathbb{R}$. This can be written explicitly as

$$\hat{H} = \begin{pmatrix} d_0 + d_3 & d_1 - id_2 \\ d_1 + id_2 & d_0 - d_3 \end{pmatrix}.$$

It is customary to define the vectors $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$, $\vec{d} = (d_1, d_2, d_3)$ and write the linear combination stated before in the more compact form

$$\hat{H} = d_0 \hat{I} + \vec{d} \cdot \vec{\sigma}.$$

Moreover, we can rewrite the eigenvalues in terms of d_0 and $\vec{d}$,

$$E_{0,1} = d_0 \pm \|\vec{d}\|,$$

meaning that our eigenvalues can be written purely in terms of the coefficients of the linear combination of the Pauli matrices.

Now, let us calculate the eigenstates associated with those eigenvalues. To do so, we will work around the problem, recalling the eigenvalue equation

$$\hat{H}|0, 1\rangle = (d_0 \hat{I} + \vec{d} \cdot \vec{\sigma})|0, 1\rangle = E_{0,1}|0, 1\rangle$$

We can see from this equation that the operator $\vec{d} \cdot \vec{\sigma}$ has the same eigenstates as $\hat{H}$, and its eigenvalues differ only by the constant d_0 , i.e., $E_{0,1} - d_0 = \pm \|\vec{d}\|$. Then, let us consider this new problem to find the eigenstates more directly.

A further simplification can be made with a change of variables $(d_1, d_2, d_3) \mapsto (r, \theta, \varphi)$ given by

$$d_1 = d \sin \theta \cos \varphi; \quad d_2 = d \sin \theta \sin \varphi; \quad d_3 = d \cos \theta,$$

where $\|\vec{d}\| = d$. So the operator $\vec{d} \cdot \vec{\sigma}$ can be written in the following way

$$\vec{d} \cdot \vec{\sigma} = \begin{pmatrix} d_3 & d_1 - id_2 \\ d_1 + id_2 & -d_3 \end{pmatrix} = d \begin{pmatrix} \cos \theta & \sin \theta e^{-i\varphi} \\ \sin \theta e^{i\varphi} & \cos \theta \end{pmatrix},$$

solving for both cases, $|0\rangle, |1\rangle$, we can find that the eigenstates are

$$|0\rangle = \begin{pmatrix} \cos(\theta/2)e^{-i\varphi} \\ \sin(\theta/2) \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} -\sin(\theta/2)e^{-i\varphi} \\ \cos(\theta/2) \end{pmatrix}$$

These eigenstates are normalized vectors, as we can see

$$\begin{aligned} \langle 0 | 0 \rangle &= (\cos(\theta/2)e^{i\varphi} \quad \sin(\theta/2)) \begin{pmatrix} \cos(\theta/2)e^{-i\varphi} \\ \sin(\theta/2) \end{pmatrix} \\ &= \cos^2(\theta/2)e^{i\varphi}e^{-i\varphi} + \sin^2(\theta/2) \\ &= 1, \end{aligned}$$

for the eigenstate $|1\rangle$ is the same case. Moreover, these eigenstates are orthogonal. Indeed,

$$\begin{aligned} \langle 0 | 1 \rangle &= (\cos(\theta/2)e^{i\varphi} \quad \sin(\theta/2)) \begin{pmatrix} \sin(\theta/2)e^{-i\varphi} \\ \cos(\theta/2) \end{pmatrix} \\ &= -\sin(\theta/2)\cos(\theta/2) + \sin(\theta/2)\cos(\theta/2) \\ &= 0. \end{aligned}$$

1.4 The Bloch sphere representation

As in the qubit case, the preceding analysis is not completely useful in many applications or appearances of two-level quantum systems. In turn, there is a simple representation, although not as insightful, named after the physicist Felix Bloch. In this representation, we represent any physically observed pure state as a point on the unit sphere. The process followed can be found in [2, see pp. 160]

Consider a two-dimensional quantum system. The two-dimensional part implies that the space of solutions, the Hilbert space associated, $\mathbb{H}$ is the vector space $\mathbb{C}^2$. Meaning that every possible state can be written as a linear combination of two linearly independent vectors. If we denote any arbitrary convenient pair of these vectors as $|0\rangle, |1\rangle \in \mathbb{C}^2$, then

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad a, b \in \mathbb{C},$$

where $|\psi\rangle \in \mathbb{C}^2$ is an arbitrary quantum state. For $|\psi\rangle$ to be of physical importance, we need it to satisfy the unit probability condition $\|a\|^2 + \|b\|^2 \leq 1$. If $|\psi\rangle$ satisfies the equality, we said that $|\psi\rangle$ is a pure state. The inequality of physically accepted states stands for the so-called mixed states [see 21, pp. 458]. In the analysis presented here, we will restrict ourselves to exploring only the pure-state case.

For instance, the pure states satisfy the four-dimensional sphere equation in $\mathbb{R}^4$. To see this explicitly, note the following

$$\|a\|^2 + \|b\|^2 = \Re(a)^2 + \Im(a)^2 + \Re(b)^2 + \Im(b)^2 = 1.$$

So, each pure state corresponds to a unique point on the four-dimensional sphere $\mathbb{S}^3$.

Furthermore, by complex analysis, we can rewrite the linear combination that gives an arbitrary pure state in the following way,

$$|\psi\rangle = r_a e^{i\varphi_a} |0\rangle + r_b e^{i\varphi_b} |1\rangle.$$

But, in quantum mechanics, the object that has all physical information is the density matrix, which for the pure state case is given by

$$\rho = |\psi\rangle\langle\psi| = e^{i\eta} |\psi\rangle e^{-i\eta} \langle\psi|,$$

where η is any real number. As you can see, any global phase factor does not change the physical information of the state, which means that $|\psi\rangle \sim e^{i\eta} |\psi\rangle$ physically. Then,

$$|\psi\rangle \sim e^{-i\varphi_a} |\psi\rangle = r_a |0\rangle + r_b e^{i(\varphi_b - \varphi_a)} |1\rangle.$$

But, $|r_a|^2 + |r_b e^{i(\varphi_b - \varphi_a)}|^2 = 1$, which can be rewritten using $z = r_a$ and $x + iy = r_b e^{i(\varphi_b - \varphi_a)}$, giving us the three-dimensional sphere equation $x^2 + y^2 + z^2 = 1$. Naturally, we can think

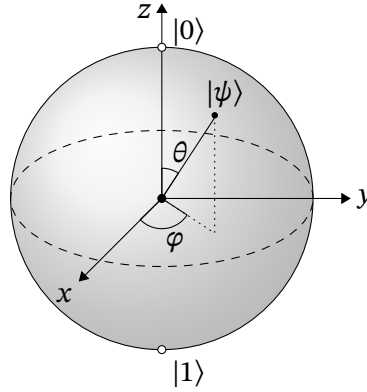


Figure 1.1: The Bloch sphere representation of a two-level quantum system.

of x, y, z as coordinates and transform them into a spherical system. Applying this idea, the transformation is given by the equations

$$x = \sin \theta \cos \varphi; \quad y = \sin \theta \sin \varphi; \quad z = \cos \theta$$

and we come up with the following equation for any pure state,

$$|\psi\rangle = \cos \theta |0\rangle + \sin \theta e^{i\varphi} |1\rangle.$$

The previous result means that any observable pure state is mathematically determined by the parameters θ, φ of the spherical coordinates. In other words, we can picture these states as points in the unitary sphere of two dimensions. This sphere is called the Bloch sphere; it provides a testbed for ideas about quantum computation and quantum information, but it is not well generalized to more than a single qubit.

Note, from Figure 1.1, that we put $|0\rangle$ in the north pole while $|1\rangle$ in the south pole. This is a well-known convention, but our current expression does not follow it,

$$|\psi(\pi, \varphi)\rangle = \cos \pi |0\rangle + \sin \pi e^{i\varphi} |1\rangle = -|0\rangle.$$

So, we need to change $\theta \rightarrow \theta/2$. Therefore, the pure states are given by

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} e^{i\varphi} |1\rangle. \quad (1.2)$$

1.5 Quantum gates

One of the most important features of information processing is that it is able to "transform" the bits to represent information. These transformations are done via electrical circuits containing

wires and logic gates. The wires are used to carry information around the circuit, while the logic gates perform manipulations of the information. The last of these has an interesting counterpart in quantum computation, the so-called quantum gates.

Let's begin analyzing the only non-trivial example of classical logic gates. This gate is called the NOT gate, which transforms the values of the bit in the following way: $0 \mapsto 1$ and $1 \mapsto 0$. The quantum case of the NOT gate must act in a similar fashion, but recall eq. 1.1,

$$|\psi\rangle = a|0\rangle + b|1\rangle.$$

We have to take into account the superposition states; to do this, we can define the quantum NOT gate as the transformation that changes $|\psi\rangle$ as follows,

$$\text{NOT}(|\psi\rangle) = a|1\rangle + b|0\rangle.$$

Note that the NOT gate returns another possible quantum state in the complex vector space; this is a crucial part of the transformations; you shouldn't be able to transform a qubit to something that is not a qubit. Therefore, this kind of quantum transformation must be an operator on the Hilbert space representing the qubit. However, these operators need to consider some properties of the possible quantum states linked to the measurement process. In fact, those operators need to be among those called unitary operators.

Definition 1.5.1 (Unitary operators). *A bounded linear operator $U : \mathbb{H} \rightarrow \mathbb{H}$, where $\mathbb{H}$ is a Hilbert space, is called unitary if it satisfies that*

$$U^\dagger U = 1 \quad \text{and} \quad UU^\dagger = 1.$$

In other words, the inverse of U satisfies that $U^{-1} = U^\dagger$.

This type of linear operator, which can also be viewed as a matrix, has a nice set of properties. The main one, and the reason to be the kind of operator that works for a quantum gate, is the preservation of the inner product. Indeed, if U is a unitary operator, then

$$\langle U\psi | U\varphi \rangle = \langle \psi | U^\dagger U \varphi \rangle = \langle \psi | \varphi \rangle.$$

Furthermore,

- The rows of U form an orthonormal basis.
- The columns of U form an orthonormal basis.
- The eigenvalues of U are all of the form $e^{i\varphi}$.

Also, in matrix form, a unitary operator (in this case, a two-dimensional one) can be diagonalized in the general form,

$$U = \begin{pmatrix} e^{i\varphi_1} & 0 \\ 0 & e^{i\varphi_2} \end{pmatrix},$$

which, geometrically, can be seen as a rotation. In fact, all quantum gates are rotations of the complex vector space associated with the qubit [3].

We should remark that the unitary condition of the operator does not ensure an efficient implementation; most of them cannot be efficiently implemented, even approximately.

The Pauli matrices and the Hadamard gate give the main examples of quantum gates in a single qubit system (a two-dimensional state space). Let us briefly describe their behavior.

As we mentioned, the quantum NOT operator changes the states $|0\rangle$ and $|1\rangle$ in any given superposition. This can be viewed as the matrix

$$\hat{\sigma}_1|\psi\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix}.$$

Therefore, the NOT gate is given by the first Pauli matrix σ_1 . Another important quantum gate is the Phase Flip operator given by the third Pauli matrix,

$$\hat{\sigma}_3|\psi\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ -b \end{pmatrix}.$$

As its name suggests, the Phase Flip operator changes the relative phase of a superposition in the standard basis. Finally, for the Pauli matrices, the second Pauli matrix is a combination of the NOT gate and the Phase Flip gate,

$$-i\hat{\sigma}_2|\psi\rangle = -i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ -a \end{pmatrix}.$$

The Hadamard gate is the last example we will mention. Its matrix form is given by

$$\mathcal{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

This gate is a particular rotation of $\pi/4$ followed by a reflection, transforming $|\psi\rangle$ such that $|0\rangle$ and $|1\rangle$ have equal probability. Also, it is a symmetric operator since it is real, i.e., $\mathcal{H}^\dagger = \mathcal{H}$ and $\mathcal{H}^2 = I$. The Hadamard Gate is one of the most important gates, with applications in quantum algorithms such as the quantum Fourier transform or some examples in quantum search algorithms.

2 | HOPF REPRESENTATION OF THE QUBIT

As it can be appreciated, the Bloch sphere gives a nice and straightforward representation of the physical and pure states of the qubit. Nonetheless, it lacks some rigor and hints at a more profound connection between the sphere $\mathbb{S}^3$ and the sphere $\mathbb{S}^2$. This chapter introduces the necessary connections, advised by the physical constraints, to find the relationship between these spheres more rigorously.

2.1 Complex projective line representation

The first step we want to take is to show that the physically observable pure states of the two-level quantum system can be viewed as points in the complex projective line. Let's introduce this new space first.

One powerful way to construct new topological spaces from a specific space that we already know is through the quotient topology. This method allows us to identify some points in a given topological space with each other to obtain a new, smaller space.

Definition 2.1.1 (Quotient topology). *Let X be a topological space, Y be any set and $q : X \rightarrow Y$ be a surjective map. We define a topology on Y by declaring that a subset $U \subseteq Y$ is open if and only if $q^{-1}(U)$ is open in X . This is called the **quotient topology** induced by the map q .*

Some basic results can be found in Appendix A; we recommend [13, pp. 65] as our primary reference or [22, pp. 136].

The application of the quotient topology that we are interested in is the quotient space. Let $\sim$ be an equivalence relation on a topological space X . For each $p \in X$, let $[p]$ denote the equivalence class of p , and let $X/\sim$ denote the set of equivalence classes. Let $q : X \rightarrow X/\sim$ be the natural projection sending each element of X to its equivalence classes. Then $X/\sim$ together with the quotient topology induced by q is called the **quotient space**. One important example is the projective space, which is of interest.

Example (Projective space): Let us define $\mathbb{P}^n$, the **projective space of dimension n** , to be the set of 1-dimensional linear subspaces (lines through the origin) of a vector space such as $\mathbb{R}^n$ or $\mathbb{C}^n$, in such cases, we call the **real or complex projective space**,

respectively. And we denote $\mathbb{RP}^n$ or $\mathbb{CP}^n$ depending on the situation. There is a natural map

$$q : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$$

defined by sending a point x to its span. We topologize the complex case $\mathbb{CP}^n$ using the quotient $(\mathbb{C}^{n+1} \setminus \{0\}/\mathbb{C}^*)$, where $\mathbb{C}^*$ is the group of nonzero complex numbers acting by scalar multiplication.

The projective space can also be viewed in another way. In the complex case, if we define an equivalence relation on $\mathbb{C}^{n+1} \setminus \{0\}$ by declaring two points z, w equivalent if $z = \lambda w$ for some nonzero complex number λ , then there is an obvious identification between $\mathbb{CP}^n$ and the set of equivalence classes. Under this identification, the map q defined above is just the map sending a point to its equivalence class.

The one-dimensional case, $\mathbb{CP}^1$, is what we call the complex projective line. Now, recall that all physically permissible (and of our interest) states had to satisfy the following properties:

$$\langle \psi | \psi \rangle = 1; \quad |\psi\rangle \sim e^{i\phi} |\psi\rangle = |\psi'\rangle.$$

If we take the physically equivalent states, we can build sets of pure states with the same observables. Each of these sets can be mapped then to more general equivalence classes,

$$\{|\psi'\rangle : |\psi'\rangle = e^{i\phi} |\psi\rangle\} \mapsto [\psi] = \{|\psi'\rangle : |\psi'\rangle = c|\psi\rangle, c \in \mathbb{C}\}$$

So, we can map the set of all pure states to the equivalence classes given by the second property. In other words, we can identify each pure state with one point in the complex projective line $\mathbb{CP}^1$.

Mathematically, this means using the quotient map,

$$q : \mathbb{C}^2 \setminus \{0\} \longrightarrow \mathbb{CP}^1.$$

Since q is a quotient map, q is surjective and continuous. Furthermore, since $\mathbb{S}^3$ is a subset of $\mathbb{C}^2 \setminus \{0\}$, we can also consider the restriction

$$E = q|_{\mathbb{S}^3} : \mathbb{S}^3 \longrightarrow \mathbb{CP}^1,$$

which is also continuous. Therefore, if we name $N : \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{S}^3$ the map that sends each vector to its normalized counterpart, we have the following commutative diagram:

$$\begin{array}{ccc} \mathbb{C}^2 \setminus \{0\} & & \\ \downarrow N & \searrow q & \\ \mathbb{S}^3 & \xrightarrow{E} & \mathbb{CP}^1 \end{array}$$

2.2 The extended complex plane & the stereographic projection

As we saw previously, the complex projective space can be identified as the space of all pure states that can be physically observed. Therefore, a critical step in our analysis is to relate $\mathbb{CP}^1$ to a new space called the extended complex plane, $\mathbb{C}_\infty$, which is closely related to the two-dimensional sphere. So, first, let us introduce this new space.

A natural way to uncover the extended complex plane is to define a famous map called stereographic projection. We will follow the process described in [23, pp. 19]. Let $N = (0, 0, 1) \in \mathbb{R}^3$ be the north pole of the sphere $\mathbb{S}^2$. Furthermore, consider the complex plane as

$$\mathbb{C} \approx \{(x_1, x_2, 0) | x_1, x_2 \in \mathbb{R}\} \subseteq \mathbb{R}^3$$

via the natural map $x_1 + ix_2 \mapsto (x_1, x_2, 0)$ and vice versa. Then, the intersection of $\mathbb{C}$ and $\mathbb{S}^2$ will lie along the equator. Now, for each point $z = x + iy \in \mathbb{C}$, consider the straight line in $\mathbb{R}^3$ that connects z and N . This will intersect the sphere at exactly one point $Z = (x_1, x_2, x_3) \neq N$, allowing us to define a map between the complex plane and the sphere.

The equation for this line in $\mathbb{R}^3$ will be $\{(N - z)t + z | -\infty < t < \infty\}$. That is, the set of all points on the line is,

$$\{(1 - t)x, (1 - t)y, t) | -\infty < t < \infty\}.$$

Hence, we can find the point Z if we can find the value of t at which this line intersects $\mathbb{S}^2$. If t is this value, then the point must satisfy the equation of the sphere, i.e.,

$$1 = (1 - t^2)x^2 + (1 - t^2)y^2 + t^2 = (1 - t^2)\|z\|^2 + t^2.$$

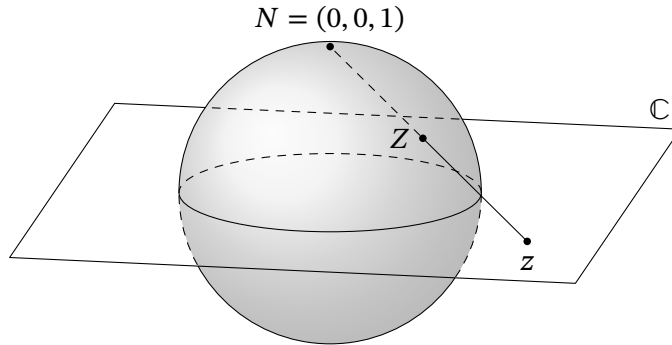


Figure 2.1: Stereographic projection. For $\|z\| > 1$, the point Z will be in the northern hemisphere; meanwhile, for $\|z\| < 1$, the point Z will be in the southern hemisphere.

Therefore, if $t \neq 1$,

$$t = \frac{\|z\|^2 - 1}{\|z\|^2 + 1}.$$

Thus, substituting in the parametric equation of the line, we get

$$x_1 = \frac{2x}{\|z\|^2 + 1}, \quad x_2 = \frac{2y}{\|z\|^2 + 1}, \quad x_3 = \frac{\|z\|^2 - 1}{\|z\|^2 + 1}. \quad (2.1)$$

Moreover, the coordinates of z can be expressed as follows

$$2x = z + \bar{z}; \quad 2y = -i(z - \bar{z}).$$

So, we now have expressions that let us find each of the coordinates of Z in terms of the associated point in the complex plane z ,

$$x_1 = \frac{z + \bar{z}}{\|z\|^2 + 1}, \quad x_2 = \frac{-i(z - \bar{z})}{\|z\|^2 + 1}, \quad x_3 = \frac{\|z\|^2 - 1}{\|z\|^2 + 1}.$$

Furthermore, we can work out the inverse map and find z in terms of the coordinates of Z . To do so, note that,

$$x_1 + ix_2 = \frac{2z}{\|z\|^2 + 1}; \quad 1 - x_3 = \frac{2}{\|z\|^2} + 1,$$

which means that z is given by

$$z = \frac{x_1 + ix_2}{1 - x_3}.$$

The map defined with our previous results $\sigma : \mathbb{S}^2 \setminus \{N\} \rightarrow \mathbb{C}$, which is written explicitly

$$\sigma(x_1, x_2, x_3) = \frac{x_1 + ix_2}{1 - x_3} \quad (2.2)$$

is the stereographic projection. This map is a homeomorphism: it is bijective, continuous with continuous inverse σ^{-1} given by

$$\sigma^{-1}(z) = \left(\frac{z + \bar{z}}{\|z\|^2 + 1}, \frac{-i(z - \bar{z})}{\|z\|^2 + 1}, \frac{\|z\|^2 - 1}{\|z\|^2 + 1} \right), \quad (2.3)$$

or in terms of the real and imaginary parts of z ,

$$\sigma^{-1}(z) = \frac{1}{x^2 + y^2 + 1} (2x, 2y, x^2 + y^2 - 1). \quad (2.4)$$

Moreover, Figure 2.1 shows that when z tends to infinity, Z tends to the north pole N . This suggests that we introduce the **point at infinity**, $z = \infty$, as the image of the north pole by the stereographic projection. The complex plane together with this point is called the **extended complex plane** $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$.

Then, we can extend the stereographic projection defining $\sigma(N) = \infty$. And this extension of the stereographic projection is also a homeomorphism that connects $\mathbb{C}_\infty$ to the entire sphere $\mathbb{S}^2$. This spherical representation is often called the **Riemann Sphere**.

When it comes to the continuity of the extended version of σ , defining a metric in $\mathbb{C}_\infty$ is helpful. One prevalent metric is the **chordal metric**, which we will derive briefly. To build the chordal metric, we define it as the distance between $z, z' \in \mathbb{C}_\infty$, $d(z, z')$, to be the distance between the corresponding points $Z, Z' \in \mathbb{S}^2 \subseteq \mathbb{R}^3$. Therefore, if $Z = (x_1, x_2, x_3)$ and $Z' = (x'_1, x'_2, x'_3)$,

$$d(z, z') = \sqrt{(x_1 - x'_1)^2 + (x_2 - x'_2)^2 + (x_3 - x'_3)^2}$$

which in turn can be written as

$$\begin{aligned} [d(z, z')]^2 &= ((x_1 - x'_1)^2 + (x_2 - x'_2)^2 + (x_3 - x'_3)^2) \\ &= \sum_{i=1}^3 (x_i^2 - 2x_i x'_i + x_i'^2). \end{aligned}$$

Using that Z, Z' are points in the sphere and the transform equations given by eq. 2.1, we obtain:

$$\begin{aligned} [d(z, z')]^2 &= 2 - 2 \left[\frac{4xx' + 4yy' + (\|z\|^2 - 1)(\|z'\|^2 - 1)}{(\|z\|^2 + 1)(\|z'\|^2 + 1)} \right] \\ &= 2 \left[\frac{(\|z\|^2 + 1)(\|z'\|^2 + 1) - 2xx' - 2yy' - (\|z\|^2 - 1)(\|z'\|^2 - 1)}{(\|z\|^2 + 1)(\|z'\|^2 + 1)} \right] \\ &= 4 \left[\frac{\|z\|^2 + \|z'\|^2 - 2xx' - 2yy'}{(\|z\|^2 + 1)(\|z'\|^2 + 1)} \right] \\ &= 4 \left[\frac{(x - x')^2 + (y - y')^2}{(\|z\|^2 + 1)(\|z'\|^2 + 1)} \right]. \end{aligned}$$

Therefore, the chordal metric (for $z, z' \neq \infty$) is given by

$$d(z, z') = \frac{2\|z - z'\|}{[(\|z\|^2 + 1)(\|z'\|^2 + 1)]^{1/2}} \quad (2.5)$$

This metric has an intriguing result for $z' = \infty$; the procedure is the same as before but using the correspondence $N = (0, 0, 1) \mapsto \infty$. Then,

$$d(z, \infty) = \frac{2}{[\|z\|^2 + 1]^{1/2}}.$$

2.3 The extended complex plane & the complex projective line

Now that we know what we will refer to as the extended complex plane or $\mathbb{C}_\infty$. Let's see how we can relate the set of observable pure states to points in this space. To do so, let us define a function we will refer to as the ratio function:

$$\zeta : \mathbb{C}^2 \longrightarrow \mathbb{C}_\infty; \quad \text{which sends } \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \longmapsto \begin{cases} \frac{\alpha}{\beta} & \text{if } \beta \neq 0; \\ \infty & \text{if } \beta = 0. \end{cases}$$

The noteworthy thing is that all elements in the equivalence classes of $\mathbb{CP}^1$ are mapped to the same point in $\mathbb{C}_\infty$,

$$c \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \mapsto \frac{c\alpha}{c\beta} = \frac{\alpha}{\beta}.$$

In other words, the complex lines that pass through the origin have the same value under ζ . So, we can rethink our ratio function as a function between $\mathbb{CP}^1$ and $\mathbb{C}_\infty$, sending the equivalence class to the value of the original map, corresponding to any element in the class. Clearly, this new function is well-defined. Furthermore, the ratio function is surjective since for any $w \in \mathbb{C}_\infty$, the equivalence class

$$\left[\begin{pmatrix} w \\ 1 \end{pmatrix} \right] \mapsto w \quad \text{or} \quad \left[\begin{pmatrix} \alpha \\ 0 \end{pmatrix} \right] \mapsto \infty, \alpha \in \mathbb{C}.$$

Also, it is injective, given α, β the entries of a vector in $\mathbb{CP}^1$. Then, for any complex numbers w_1, w_2 , we have that $w_1 = a\alpha$ and $w_2 = b\beta$, with $a, b \in \mathbb{C} \setminus \{0\}$. Moreover, if

$$\left[\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \right] = \left[\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \right] \quad \text{then} \quad \left(\frac{a}{b} \right) \left(\frac{\alpha}{\beta} \right) = \frac{w_1}{w_2} = \frac{\alpha}{\beta},$$

so $a = b$. If $\beta = 0$, then if the equivalence classes are the same, we must have that $w_2 = 0$, so there is no restriction on α . Therefore, the ratio map is bijective and continuous. Moreover, the inverse function,

$$\tilde{\zeta} : \mathbb{C}_\infty \longrightarrow \mathbb{CP}^1; \text{ which sends } z \mapsto \begin{cases} \begin{pmatrix} z \\ 1 \end{pmatrix} & \text{if } z \in \mathbb{C}; \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \text{if } z = \infty. \end{cases}$$

its also continuous. So, it is a homeomorphism.

If we recall the result given in eq. 1.2 about the Bloch sphere representation, we have that for a pure state,

$$\zeta = e^{-i\varphi} \cot \frac{\theta}{2}.$$

Another thought-provoking insight the ratio map gives us is the correspondence of unitary operators in the extended complex plane. Let U be a unitary matrix. Then it transforms the quantum state $|\psi\rangle$ as follows:

$$U|\psi\rangle = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{pmatrix}$$

Then, applying the ratio function, we get

$$\zeta(U|\psi\rangle) = \frac{a\alpha + b\beta}{c\alpha + d\beta} = \frac{a\zeta' + b}{c\zeta' + d} = f(\zeta'),$$

where $\zeta' = \alpha/\beta$ and $f : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ and the next convention holds

$$f(-\frac{d}{c}) = \infty; \quad f(\infty) = \frac{a}{c}.$$

This specific form that takes the composition of the unitary operator and ζ is important in mathematics.

Definition 2.3.1 (Linear fractional transformation). *A mapping of the form*

$$S(w) = \frac{aw + b}{cw + d}, \text{ with } w \in \mathbb{C}_\infty$$

*is called a linear fractional transformation. If a, b, c, d also satisfy $ad - bc \neq 0$, then $S(w)$ is called a **Möbius transformation**.*

The case of $\zeta \circ U$ is the particular case of a Möbius transformation, where the existence of the inverse matrix of U gave us that $\det U = ad - bc \neq 0$. The set consisting of this kind of transformation has the algebraic structure of a group, and it is called the Möbius group in the case of $\mathbb{C}_\infty$ or the one-dimensional projective group in $\mathbb{CP}^1$.

Moreover, the Möbius transformation is a conformal mapping, which means preserving angles, although not necessarily lengths. And they have the following nice property of

■ **Theorem 2.3.1.** *A Möbius transformation carries circles into circles*

Also, the Möbius transformations can be viewed as a combination of the following three specific cases of transformations. If $S(w) = w + a$ then S is called a translation; if $S(w) = aw$ with $a \neq 0$ then S is a dilation; if $S(w) = e^{i\theta}w$ then it is a rotation; finally, if $S(w) = 1/w$ it is the inversion.

□ **Proposition 2.3.1.** *A Möbius transformation is the composition of translations, dilations, and inversion.*

For example, the inversion corresponds to the NOT gate in the context of quantum gates. Some correspondences are in Table 2.1.

For a more in-depth analysis of some of the properties and their proofs of linear fractional transformations, we suggest [25, pp. 80] or [23, pp. 47].

Unitary Gate U	I	σ_1	σ_2	σ_3	H
Conformal mapping $f(\zeta)$	ζ	$1/\zeta$	$-1/\zeta$	$-\zeta$	$\frac{\zeta+1}{\zeta-1}$

Table 2.1: The correspondence between unitary operations and conformal maps [24].

2.4 The Hopf map

With the maps we have developed in the previous sections, we can build a function that connects $\mathbb{S}^3$ with $\mathbb{S}^2$ that is surjective and continuous. We will denote such a map H and is given by

$$\begin{array}{ccccccc} \mathbb{S}^3 & \xrightarrow{E} & \mathbb{CP}^1 & \xrightarrow{\zeta} & \mathbb{C}_\infty & \xrightarrow{\sigma^{-1}} & \mathbb{S}^2 \\ & & & & & \nearrow & \\ & & & & & H & \end{array}$$

Let's work out the details so we can state explicitly H . To begin with, to have the original Hopf map (the one proposed by Heinz Hopf, see [12, pp. 654]), we will take ζ as the conjugate of the original form, i.e.,

$$\zeta \left[\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \right] = \overline{\begin{pmatrix} \alpha \\ \beta \end{pmatrix}} = \frac{\bar{\alpha}\beta}{\|\beta\|^2}.$$

If we consider $\alpha = x_1 + ix_2$ and $\beta = x_3 + ix_4$, then ζ is given by

$$\zeta \begin{pmatrix} x_1 + ix_2 \\ x_3 + ix_4 \end{pmatrix} = \frac{(x_1 - ix_2)(x_3 + ix_4)}{x_3^2 + x_4^2} = \frac{(x_1x_3 + x_2x_4) + i(x_1x_4 - x_2x_3)}{x_3^2 + x_4^2}$$

Note that the x, y needed for σ^{-1} are

$$\begin{aligned} x &= \frac{x_1x_3 + x_2x_4}{x_3^2 + x_4^2} \\ y &= \frac{x_1x_4 - x_2x_3}{x_3^2 + x_4^2} \end{aligned}$$

Then, carrying out some of the calculations before applying σ^{-1} ,

$$x^2 + y^2 = \frac{(x_1x_3 + x_2x_4)^2 + (x_1x_4 - x_2x_3)^2}{(x_3^2 + x_4^2)^2} = \frac{(x_1^2 + x_2^2)}{(x_3^2 + x_4^2)},$$

and

$$\begin{aligned} x^2 + y^2 + 1 &= \frac{(x_1^2 + x_2^2)}{(x_3^2 + x_4^2)} + 1 = \frac{1}{(x_3^2 + x_4^2)} \\ x^2 + y^2 - 1 &= \frac{(x_1^2 + x_2^2)}{(x_3^2 + x_4^2)} - 1 = \frac{x_1^2 + x_2^2 - x_3^2 - x_4^2}{(x_3^2 + x_4^2)} \end{aligned}$$

Therefore, applying σ^{-1} we obtain

$$\sigma^{-1}\left(\zeta\left(E\begin{pmatrix} x_1 + ix_2 \\ x_3 + ix_4 \end{pmatrix}\right)\right) = (2(x_1x_3 + x_2x_4), 2(x_1x_4 - x_2x_3), x_1^2 + x_2^2 - x_3^2 - x_4^2),$$

then H is given explicitly by

$$H\begin{pmatrix} x_1 + ix_2 \\ x_3 + ix_4 \end{pmatrix} = (2(x_1x_3 + x_2x_4), 2(x_1x_4 - x_2x_3), x_1^2 + x_2^2 - x_3^2 - x_4^2). \quad (2.6)$$

Or, in terms of the complex coordinates of the vector in $\mathbb{S}^3$ as,

$$H\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = (2\Re(\bar{z}_1z_2), 2\Im(\bar{z}_1z_2), \|z_1\|^2 - \|z_2\|^2). \quad (2.7)$$

Also, in the case that we take $\mathbb{R}^3$ as $\mathbb{C} \times \mathbb{R}$,

$$H\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = (2\bar{z}_1z_2, \|z_1\|^2 - \|z_2\|^2).$$

But another fascinating way to write this map is using the Pauli matrices. For $|\psi\rangle = (a, b) \in \mathbb{S}^3$, consider the following expected values

$$\begin{aligned} \langle\psi|\sigma_1|\psi\rangle &= \bar{a}b + a\bar{b} \\ \langle\psi|\sigma_2|\psi\rangle &= -i\bar{a}b + ia\bar{b} \\ \langle\psi|\sigma_3|\psi\rangle &= \|a\|^2 - \|b\|^2 \end{aligned}$$

But, $2\Re(\bar{a}b) = \bar{a}b + a\bar{b}$ and $2\Im(\bar{a}b) = -i\bar{a}b + ia\bar{b}$. Therefore, the map H can be rewritten in the next form

$$H(|\psi\rangle) = (\langle\psi|\sigma_1|\psi\rangle, \langle\psi|\sigma_2|\psi\rangle, \langle\psi|\sigma_3|\psi\rangle),$$

or using the density operator $\rho = |\psi\rangle\langle\psi|$ notation,

$$H(|\psi\rangle) = (\text{Tr}(\sigma_1\rho), \text{Tr}(\sigma_2\rho), \text{Tr}(\sigma_3\rho)).$$

As we mentioned, this map is called the Hopf map and has fascinating properties, such as being continuous and onto, because E is surjective. Furthermore, an interesting fact comes when we apply the Hopf map, as in Eq. 2.7, to the Bloch representation we gave in Eq. 1.2:

$$H\begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\varphi} \end{pmatrix} = (2\cos(\theta/2)\sin(\theta/2)\cos\varphi, 2\cos(\theta/2)\sin(\theta/2)\sin\varphi, \cos^2(\theta/2) - \sin^2(\theta/2)).$$

Using the fact that $2\cos(\theta/2)\sin(\theta/2) = \sin\theta$ and $\cos^2(\theta/2) - \sin^2(\theta/2) = \cos\theta$, we have that

$$H\begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\varphi} \end{pmatrix} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta),$$

which are the ordinary spherical coordinates we know. Not only that, but it also has the exciting feature of being a fiber and principal bundle. Let us introduce these concepts in the following sections.

2.5 Topological groups

Before discussing bundles, let us introduce the concept of a topological group and some of its properties. This section will briefly describe definitions for later use; for more in-depth about groups, you can consult [26] and [27]. Let us begin to define what we understand as a group.

Definition 2.5.1 (Group). *A group is a non-empty set G together with a binary operation on G , which is*

- *Associative: $a(bc) = (ab)c$ for all $a, b, c \in G$.*
- *There exists an identity element $e \in G$ such that $ae = ea = a$ for all $a \in G$.*
- *For all $a \in G$, there exists an inverse element $a^{-1} \in G$ such that $a^{-1}a = aa^{-1} = e$.*

This algebraic structure is particularly useful if one considers it a set of permutations of some arbitrary set X (which can be a topological space). In fact, the set of possible permutations or bijections from X to X form a group called the symmetric group of X , denoted by S_X . Furthermore, if a homomorphism exists that allows us to think of a particular group G as S_X , we can use the elements of G to find new values of an element $x \in X$. The following definition gives this abstraction:

Definition 2.5.2 (Action of a group G). *If X is a set and G is a group, then X is a G -set if there is a function $\alpha : X \times G \rightarrow X$ (called a right **action**), denoted by $\alpha : (x, g) \mapsto xg$, such that:*

- *$xe = x$ for all $x \in X$.*
- *$(xh)g = x(hg)$ for all $g, h \in G$ and $x \in X$.*

One also says that G acts on X . One can define a left action similarly but with $\alpha : G \times X \rightarrow X$.

Suppose X is a topological space and a G -set; the action on X is called an **action by homeomorphisms** if, for each $g \in G$, the map $x \mapsto xg$ is a homeomorphism of X . Moreover, the action is said to be continuous if the map $G \times X \rightarrow X$ is continuous. Moreover, we can separate X into subsets via the action of G on X .

Definition 2.5.3. *If X is a G -set and $x \in X$, then the G -orbit or just the **orbit** of x is*

$$xG = \{xg : g \in G\} \subseteq X.$$

The orbits of X form a partition via an equivalence relation given by $x \equiv y$ with $x, y \in X$ if $y = xg$ for some $g \in G$. Also, if X is a topological space, we can build a new topological space via the quotient topology, and X/G denotes it. Now, we are ready to define what a topological group is.

Definition 2.5.4 (Topological group). A topological group G is a set G with the structure of a group given by Definition 2.5.1 and a topological space (see Definition A.1.1) with a topology such that the maps $m : G \times G \rightarrow G$ and $i : G \rightarrow G$ given by

$$m(g, h) = gh, \quad i(g) = g^{-1}$$

are continuous. The continuity of m is understood to be with respect to the product topology of $G \times G$. If G is acting on a space X , we said it is a **G -space** instead of a G -set.

Some subsets of matrices have the structure of a group and are topological groups. That is the case of the nonsingular matrices $GL(n, \mathbb{R})$ and $GL(n, \mathbb{C})$ and the case of the orthogonal and unitary matrices, $O(n)$, $SO(n)$, $U(n)$, $SU(n)$.

 **Proposition 2.5.1.** For a G -space X , the map $x \mapsto xg$ is a homeomorphism, and then the quotient map $\mu : X \rightarrow X/G$ is open.

Proof. The inverse of $x \mapsto xg$ is the map $x \mapsto xg^{-1}$, and is continuous by the definition of G as a topological group.

Now, for μ , consider an open set $W \subseteq X$. Recall that μ is a quotient map and, therefore, is continuous. Then, if $\mu^{-1}(\mu(W))$ is open, $\mu(W)$ will be open. Indeed,

$$\mu^{-1}(\mu(W)) = \bigcup_{g \in G} Wg,$$

which is open because each Wg is an open set by the previous result. Therefore, $\mu(W)$ is open, and μ is an open map. □

Definition 2.5.5 (Free G -space). A G -space X is called free, provided that the action of G on X has the property that $xg = x$ if and only if $g = e$.

2.6 Fiber and principal bundles

We will first define what a bundle is to give you some intuition of the principal and fiber bundle concepts.

Definition 2.6.1 (Bundle). A bundle is a triple (E, p, B) , where $p : E \rightarrow B$ is a map. The space B is called the **base space**, the space E is called the **total space**, and the map p is called the **projection of the bundle**. For each $b \in B$, the space $p^{-1}(b)$ is called the **fibre** of the bundle over $b \in B$.

Intuitively, one thinks of a bundle as a union of fibers $p^{-1}(b)$ for $b \in B$ parametrized by B and “glued together” by the topology of the space E . But we can go further with the structure using a topological group G . In such a case, the bundle is called a principal or fiber bundle. Roughly speaking, a fiber bundle E on a base B with fiber F is a geometric way to express E in terms of B and F . That is, locally, we will express E as the product of B and F . In the principal bundle case, the fiber F is a topological group.

We will start defining a principal bundle to better understand the more complex case of the fiber bundle.

Definition 2.6.2 (Principal G -bundle). Let G be a topological group. A principal G -bundle consists of a continuous function $p : P \rightarrow X$ and a continuous right action $P \times G \rightarrow P$ such that there is an open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ of X together with homeomorphisms $\psi_\alpha : U_\alpha \times G \rightarrow p^{-1}(U_\alpha)$ that satisfy:

- The following diagram is commutative:

$$\begin{array}{ccc} U_\alpha \times G & \xrightarrow{\psi_\alpha} & p^{-1}(U_\alpha) \\ & \searrow \pi & \downarrow p| \\ & & U_\alpha \end{array}$$

- $\psi_\alpha(x, g) = \psi_\alpha(x, e)g$ with the identity $e \in G$ and $g \in G$.

The open cover and the homeomorphisms ψ_α are called the **local trivialization** and are generally not unique.

A simple example is the product principal G bundle or simply the trivial principal bundle.

Example (The trivial principal bundle): The trivial G -principal bundle in the space X consists of a continuous function $p : X \times G \rightarrow X$ such that $p(x, g) = x$, an action $(X \times G) \times G \rightarrow X \times G$ such that $(x, g)h = (x, gh)$ for $h \in G$ and an open cover $\mathcal{U} = \{X\}$. Obtaining,

$$\begin{array}{ccc} X \times G & \xrightarrow{\psi=Id} & X \times G \\ & \searrow \pi_X & \downarrow p \\ & & X \end{array}$$

which satisfies the first condition of the definition. Moreover,

$$\psi(x, e)g = (x, e)g = (x, eg) = (x, g) = \psi(x, g).$$

Satisfying the second condition. Therefore, we see that the trivial G -principal bundle is, in fact, a G -principal bundle.

Let us define some supplementary functions that will allow us to prove some of the key features of principal G -bundles.

Definition 2.6.3. Let $p : P \rightarrow X$ be a principal G -bundle with local trivialization $\{U_\alpha\}$ and ψ_α . Also, consider $x \in U_\alpha$. We define the function,

$$\psi_{\alpha,x} : G \rightarrow p^{-1}(x)$$

as $\psi_{\alpha,x}(g) = \psi_\alpha(x, g)$.

The past definition makes sense since

$$p|_{p^{-1}(U_\alpha)}(\psi_\alpha(x, g)) = \pi(x, g) = x;$$

meaning that $\psi_\alpha(x, g) \in p^{-1}(x)$. Furthermore, the function $\psi_{\alpha,x}$ defines a homeomorphism. Indeed, consider the inverse function of ψ_α as $\hat{\psi}_\alpha : p^{-1}(U_\alpha) \rightarrow U_\alpha \times G$. The inverse function satisfies that

$$\pi \circ \hat{\psi}_\alpha = p|_{p^{-1}(U_\alpha)} \circ \psi_\alpha \circ \hat{\psi}_\alpha = p|_{p^{-1}(U_\alpha)},$$

which let us define the inverse function $\hat{\psi}_{\alpha,x} : p^{-1}(x) \rightarrow G$.

Moreover, the homeomorphisms $\psi_{\alpha,x}$ are G -equivariant, i.e., $\psi_{\alpha,x}(gh) = \psi_{\alpha,x}(g)h$. Indeed,

$$\psi_{\alpha,x}(gh) = \psi_\alpha(x, gh) = \psi_\alpha(x, e)gh = (\psi_\alpha(x, e)g)h = \psi_\alpha(x, g)h.$$

Proposition 2.6.1. Let $p : P \rightarrow X$ be a G -principal bundle and $a \in P$ which $p(a) = x$. Then, $\iota_a : G \rightarrow p^{-1}(x)$ given by $\iota_a(g) = ag$ is a homeomorphism.

Proof. Consider $x \in U_\alpha$ and $\psi_{\alpha,x} : G \rightarrow p^{-1}(x)$ for some $\alpha \in I$. Also, let us define the left translation $\iota_{g_0} : G \rightarrow G$ given by $\iota_{g_0}(g) = g_0g$, where we denote $a = \psi_{\alpha,x}(g_0) \in P$. But, ι_{g_0} is continuous because G is a topological group. Furthermore, it is a homeomorphism since $\iota_{g_0}^{-1}$ is also a continuous function, and it is the inverse of ι_{g_0} . Additionally,

$$\psi_{\alpha,x}(\iota_{g_0}(g)) = \psi_{\alpha,x}(g_0g) = \psi_{\alpha,x}(g_0)g = ag = \iota_a(g),$$

for all $g \in G$. Therefore, $\psi_{\alpha,x} \circ \iota_{g_0} = \iota_a$, so it is a homeomorphism.

$$\begin{array}{ccc} G & \xrightarrow{\iota_a} & p^{-1}(x) \\ \iota_{g_0} \downarrow & \nearrow \psi_{\alpha,x} & \\ G & & \end{array}$$

□

▣ **Corollary 2.6.1.** *Let $p : P \rightarrow X$ be a principal G -bundle. Therefore, given $a_1, a_2 \in P$, the equation $p(a_1) = p(a_2)$ is satisfied if and only if it exists $g \in G$ such that $a_1g = a_2$.*

Proof.

($\Rightarrow$) Let $x = p(a_1) = p(a_2)$ and consider the function $\iota_{a_1} : G \rightarrow p^{-1}(x)$. Then, given that $a_2 \in p^{-1}(x)$ it exists $g \in G$ such that $a_2 = \iota_{a_1}(g) = a_1g$.

($\Leftarrow$) Let $a \in P$, $g \in G$, and $x = p(a)$. Then, $\iota_a(g) = ag \in p^{-1}(x)$ and $p(ag) = x = p(a)$. If $a_1g = a_2$, it follows that $p(a_1) = p(a_1g) = p(a_2)$. □

An additional important property for a critical result of principle bundles is stated in the following lemma. Although just half of the proof is presented, the other part is not challenging to see [28].

▣ **Lemma 2.6.1.** *Let $p : P \rightarrow X$ be a principal G -bundle. Then, p is an open and surjective map, i.e., a quotient map.*

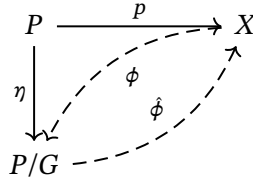
Proof. To see that p is surjective, we can proceed by contradiction. Suppose that p is not surjective, then exists $x \in X$ such that $p(a) \neq x$ for all $a \in P$. But, $\{U_\alpha\}_{\alpha \in I}$ is an open cover, so it exists U_α such that $x \in U_\alpha$. Now, consider $(x, g) \in U_\alpha \times G$ for some $g \in G$. Then,

$$x = \pi(x, g) = p|_{p^{-1}(U_\alpha)}(\psi_\alpha(x, g)) = p(\psi_\alpha(x, g)).$$

Therefore, we have our contradiction, and p is surjective. □

▣ **Corollary 2.6.2.** *Let $p : P \rightarrow X$ be a principal G -bundle. Then $X \cong P/G$.*

Proof. By Corollary 2.6.1 if $p(a_1) = p(a_2) \iff$ exists $g \in G$ such that $a_1g = a_2 \iff$ the orbit $a_1G = a_2G \iff \eta(a_1) = \eta(a_2)$. Moreover, because p and η are quotient maps, by Theorem A.3.3, there exists a homeomorphism ϕ with inverse $\hat{\phi}$ between X and P/G .



Our last result of principal bundles tells us about the algebraic structure of the topological group G that defines it. □

■ **Corollary 2.6.3.** Let $p : P \rightarrow X$ be a principal G -bundle. Then, the action P is a free G -space.

Proof. Let be $a \in P$ and $g \in G$, with $x = p(a)$. And suppose that $ag = a$, then

$$\iota_a(g) = ag = a = ae = \iota_a(e).$$

But ι is an injective function, so $g = e$. □

Now, let us define what we will understand as a fiber bundle

Definition 2.6.4 (Fiber bundle). A fiber bundle with fiber a space F consists of a continuous function $q : E \rightarrow X$ with the following property: There exists an open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ and homeomorphisms ψ_α such that the following triangle commutes

$$\begin{array}{ccc} U_\alpha \times F & \xrightarrow{\psi_\alpha} & q^{-1}(U_\alpha) \\ \pi \downarrow & \swarrow q| & \\ U_\alpha & & \end{array}$$

The primary example of a fiber bundle is the product fiber bundle given as follows:

Example (Product fiber bundle): The product fiber bundle with fiber F over X is $q : X \times F \rightarrow X$ with $\mathcal{U} = \{X\}$ and with the canonical local trivialization $Id : X \times F \rightarrow X \times F$.

In general, any principal G -bundle is a fiber bundle with fiber G . To see it, consider the following development: if $p : P \rightarrow X$ is a principal G -bundle, F is an arbitrary space, and $G \times F \rightarrow F$ is a continuous action on F by G . We define the fiber bundle associated as follows. Consider $P \times F$ with the following action

$$(P \times F) \times G \rightarrow P \times F \quad \text{given by} \quad (a, v)g = (ag, g^{-1}v);$$

where $ag \in P$ and $g^{-1}v \in F$. We will denote the space of orbits by $P \times_G F$. So, the function

$$\tilde{p} : P \times_G F \rightarrow X \quad \text{given by} \quad \tilde{p}[a, v] = p(a),$$

defines a fiber bundle. This is stated by the following proposition that we shall not prove (see [29]).

■ **Proposition 2.6.2.** Let $p : P \rightarrow X$ be a principal G -bundle and $G \times F \rightarrow F$ a continuous action. Then, $\tilde{p} : P \times_G F \rightarrow X$ is a fiber bundle with fiber F and structural group G .

2.7 The Hopf map as a fiber bundle

Now that we know what a fiber and principal bundle are, we can formally show that the Hopf map defines one. To do so, first, we will consider an alternative map h built similarly to H but not using the conjugate of the map ζ . This new map is equivalent to the one we derive, so the following proof holds in the conclusion for H . Therefore, considering $h : \mathbb{S}^3 \rightarrow \mathbb{S}^2$, we can define a principal G -bundle on $\mathbb{S}^3$ if we consider the unitary group,

$$U(1) = \{z \in \mathbb{C} : \|z\| = 1\} = \mathbb{S}^1$$

acting via the complex multiplication in each entry of the vectors on $\mathbb{S}^3$. So, we are left to define a local trivialization such that the following diagram commutes:

$$\begin{array}{ccc} U_1 \times \mathbb{S}^1 \subset \mathbb{C}_\infty \times \mathbb{S}^1 & \xrightarrow{\psi_1} & h^{-1}(U_1) \subset \mathbb{S}^3 \\ & \searrow \pi & \downarrow h| \\ & & U_1 \subset \mathbb{C}_\infty \end{array}$$

Consider U to be the open subset given by $U = \sigma(\mathbb{S}^2 \setminus \{N\}) \subset \mathbb{C}_\infty$. So, $h^{-1}(U)$ is an open subset of $\mathbb{S}^3$. We want to define the function ψ_1 such that it is a homeomorphism and makes the triangle commute. Set ψ_1 as the function given by

$$\psi_1 \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right) = \frac{1}{\sqrt{1 + \|\zeta\|^2}} \begin{pmatrix} e^{i\theta} \zeta \\ e^{i\theta} \end{pmatrix}.$$

Then, it is continuous and also has an inverse function given by

$$\hat{\psi}_1 \left(\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right) = \begin{pmatrix} z_1/z_2 \\ e^{i\varphi_2} \end{pmatrix}$$

where we use the polar form of the coordinates, $z_1 = r_1 e^{i\varphi_1}$ and $z_2 = r_2 e^{i\varphi_2}$. This function is well defined since $z_2 \neq 0$.

Before we continue, let us prove that $\hat{\psi}_1$ is indeed the inverse function of ψ_1 . Therefore,

$$\psi_1 \circ \hat{\psi}_1 \left(\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right) = \psi_1 \left(\begin{pmatrix} z_1/z_2 \\ e^{i\varphi_2} \end{pmatrix} \right) = \frac{1}{\sqrt{1 + \|z_1/z_2\|^2}} \begin{pmatrix} z_1/z_2 e^{i\varphi_2} \\ e^{i\varphi_2} \end{pmatrix} = r_2 \begin{pmatrix} r_1/r_2 e^{i\varphi_1} \\ e^{i\varphi_2} \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}.$$

The other way around also gives the desired result,

$$\hat{\psi}_1 \circ \psi_1 \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right) = \hat{\psi}_1 \left(\frac{1}{\sqrt{1 + \|\zeta\|^2}} \begin{pmatrix} e^{i\theta} \zeta \\ e^{i\theta} \end{pmatrix} \right) = \begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix}$$

Then, ψ_1 is a homeomorphism since $\hat{\psi}_1$ is also a continuous function. We are left to prove that $h \circ \psi_1 = \pi$. Indeed,

$$\begin{aligned} h \circ \psi_1 \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right) &= h \left(\frac{1}{\sqrt{1 + \|\zeta\|^2}} \begin{pmatrix} e^{i\theta}\zeta \\ e^{i\theta} \end{pmatrix} \right) = \zeta \circ E \left(\frac{1}{\sqrt{1 + \|\zeta\|^2}} \begin{pmatrix} e^{i\theta}\zeta \\ e^{i\theta} \end{pmatrix} \right) \\ &= \zeta \left(\frac{1}{\sqrt{1 + \|\zeta\|^2}} \begin{pmatrix} e^{i\theta}\zeta \\ e^{i\theta} \end{pmatrix} \right) \\ &= \zeta \\ &= \pi \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right). \end{aligned}$$

Now, consider $U_2 \subset \mathbb{C}_\infty$ be the open subset given by $U_2 = \sigma(\mathbb{S}^2 \setminus \{S = (0, 0, 1)\})$. Then, again, $h^{-1}(U_2)$ will be an open subset of $\mathbb{S}^3$. So, we want to define a homeomorphism ψ_2 such that the following diagram commutes:

$$\begin{array}{ccc} U_2 \times \mathbb{S}^1 \subset \mathbb{C}_\infty \times \mathbb{S}^1 & \xrightarrow{\psi_2} & h^{-1}(U_2) \subset \mathbb{S}^3 \\ & \searrow \pi & \downarrow h| \\ & & U_2 \subset \mathbb{C}_\infty \end{array}$$

Let's set the map

$$\psi_2 \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right) = \frac{1}{\sqrt{1 + \|\zeta^{-1}\|^2}} \begin{pmatrix} e^{i\theta} \\ e^{i\theta}\zeta^{-1} \end{pmatrix},$$

which is continuous and has an inverse given by

$$\hat{\psi}_2 \left(\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right) = \begin{pmatrix} z_1/z_2 \\ e^{i\varphi_1} \end{pmatrix}.$$

Note that it is well-defined since $z_1 \neq 0$. Let's prove that they are the inverse function of each other,

$$\psi_2 \circ \hat{\psi}_2 \left(\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right) = \psi_2 \left(\begin{pmatrix} z_1/z_2 \\ e^{i\varphi_1} \end{pmatrix} \right) = \frac{1}{\sqrt{1 + \|z_2/z_1\|^2}} \begin{pmatrix} e^{i\varphi_1} \\ e^{i\varphi_1} \frac{z_2}{z_1} \end{pmatrix} = r_1 \begin{pmatrix} e^{i\varphi_1} \\ \frac{r_2}{r_1} e^{i\varphi_2} \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}.$$

The other way around also gives the desired result,

$$\hat{\psi}_2 \circ \psi_2 \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right) = \hat{\psi}_2 \left(\frac{1}{\sqrt{1 + \|\zeta\|^2}} \begin{pmatrix} e^{i\theta} \\ e^{i\theta}\zeta^{-1} \end{pmatrix} \right) = \begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix}.$$

Then, ψ_2 is a homeomorphism since $\hat{\psi}_2$ is also a continuous function. For last, let's see if ψ_2 satisfies that $\pi = h \circ \psi_2$. Indeed,

$$\begin{aligned} h \circ \psi_2 \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right) &= h \left(\frac{1}{\sqrt{1 + \|\zeta^{-1}\|^2}} \begin{pmatrix} e^{i\theta} \\ e^{i\theta} \zeta^{-1} \end{pmatrix} \right) = \zeta \circ E \left(\frac{1}{\sqrt{1 + \|\zeta^{-1}\|^2}} \begin{pmatrix} e^{i\theta} \\ e^{i\theta} \zeta^{-1} \end{pmatrix} \right) \\ &= \zeta \left(\frac{1}{\sqrt{1 + \|\zeta^{-1}\|^2}} \begin{pmatrix} e^{i\theta} \\ e^{i\theta} \zeta^{-1} \end{pmatrix} \right) \\ &= \zeta \\ &= \pi \left(\begin{pmatrix} \zeta \\ e^{i\theta} \end{pmatrix} \right). \end{aligned}$$

Thus, we have shown that there is a principal $\mathbb{S}^1$ -bundle over $\mathbb{S}^3$ since we prove that $\mathbb{S}^3$ is locally homeomorphic to $\mathbb{C}_\infty \times \mathbb{S}^1$, i.e., $\mathbb{S}^2 \times \mathbb{S}^1$. Note that each point $h(a)$ with $a \in \mathbb{S}^3$ is a physically observed qubit state; also, two states are equal if and only if they differ by a global phase $e^{i\theta}$ (see Corollary 2.6.1). Furthermore, each point of $\mathbb{S}^2$ is a qubit state since it is homeomorphic to the space of orbits $\mathbb{S}^3/\mathbb{S}^1$, and each orbit is a qubit state (see Corollary 2.6.2). Furthermore, we have defined a fiber bundle on $\mathbb{S}^3$ with fiber $\mathbb{S}^1$ (see Proposition 2.6.2).

2.8 Visualization of the Hopf fibration.

To visualize the Hopf fibration, we can generalize the stereographic projection to project the points of the four-dimensional space to three-dimensional space. In this way, we can project the fibers $H^{-1}(a)$ or the qubit state in the three-dimensional space. This representation looks like the following figures.

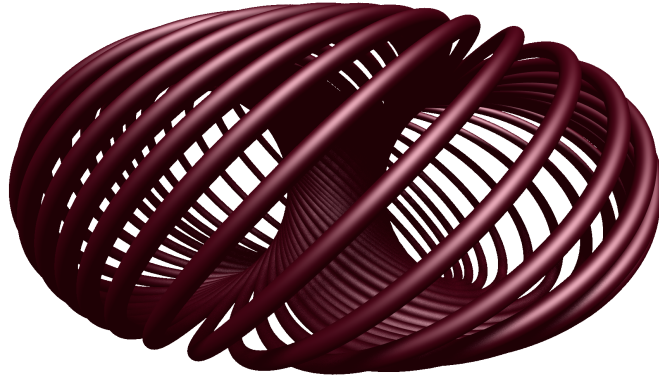


Figure 2.2: Graphical representation of a subset of points in the intersection of $\mathbb{S}^2$ and the xy -plane.

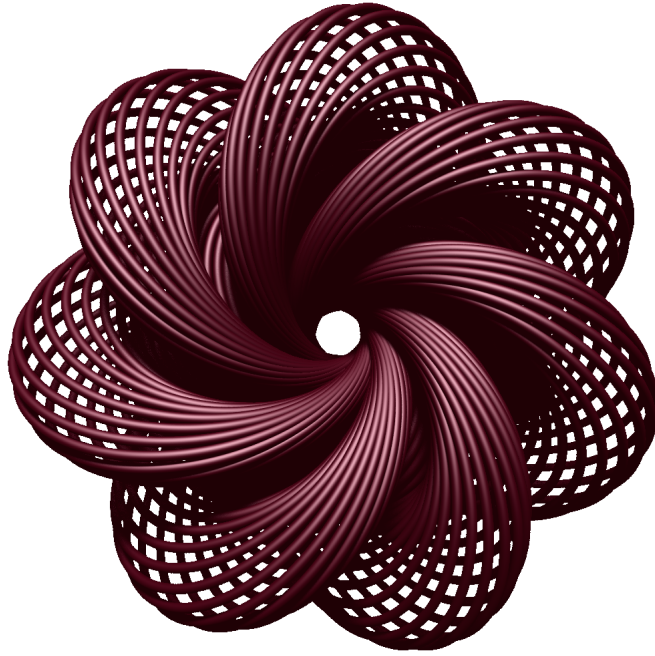


Figure 2.3: Graphical representation of a set of points of $\mathbb{S}^2$ given by the parametric functions $\varphi(t) = -(\pi/7) \sin(14\pi t) + \pi/2$ and $\theta(t) = 2\pi t + 1/2 \cos(14\pi t)$ for the spherical angles.

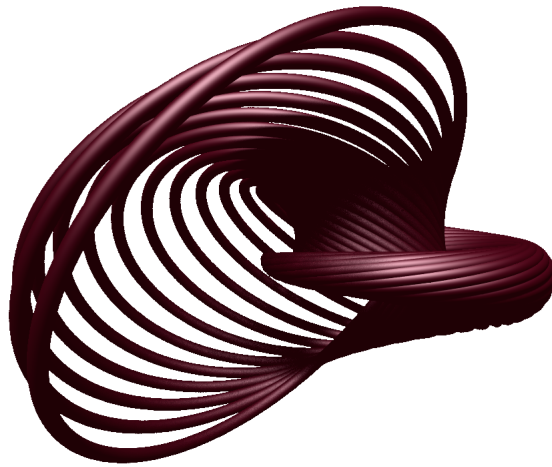


Figure 2.4: Graphical representation of a set of points of $\mathbb{S}^2$ given by equal increments of both angles θ and φ .

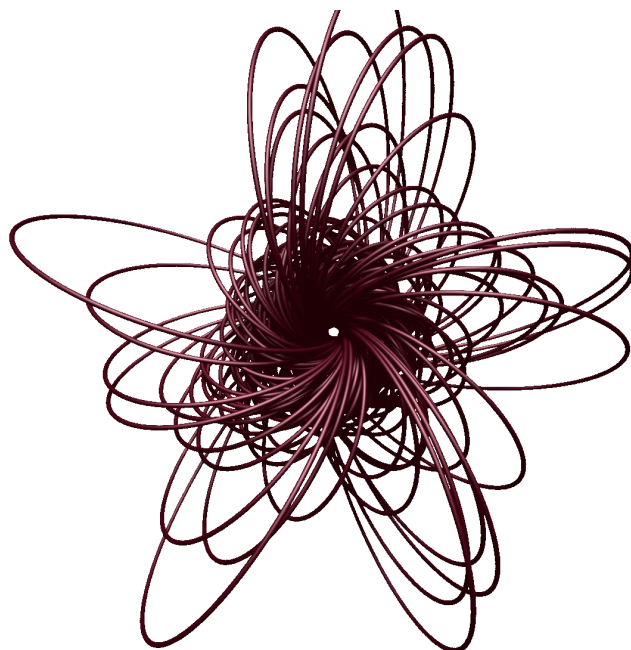


Figure 2.5: Graphical representation of a set of random points of $\mathbb{S}^2$.

3 | CONCLUSION

Throughout this work, we have witnessed the remarkable power of physical intuition. We began with a brief introduction to the concept of a qubit from a quantum information point of view. Then, we suggested that such a concept would ideally be model with a general 2-level system from quantum physics, which we shortly review from the perspective of physics. From the general solution of the eigenvalue problem to the (not rigorous) geometrical general solution, namely the Bloch sphere representation. We closed the first chapter by showcasing the four most important transformations of the qubit in the context of quantum information. Next, suggested by the Bloch representation, we followed one possible way to use the physical constraints and rigorously connect the physical solution space of the problem (the 3-dimensional sphere) to the 2-dimensional sphere. To do so, we apply the constraints to get $\mathbb{S}^3$ which we could identify to a new space, which was a specific case of the projective space. In order to connect $\mathbb{S}^2$ to the projective space, we define the extended complex plane through the (extended) stereographic projection and build a homeomorphism between them (the ratio map). The composition of all these maps gave us a historically important map on algebraic topology, namely the Hopf map, which suggested the construction of a principal bundle. To wrap up, it is introduced the general definition of a principal bundle and a proof that the Hopf map let define one, to lastly showcased some interesting visualizations of its fibers.

As mentioned before, the restriction of physical degrees of freedom by the properties of observables in the quantum mechanics of the qubit leads to the idea of identifying the physical states to different spaces from $\mathbb{S}^3$ to a 2-dimensional sphere $\mathbb{S}^2$. And, therefore, allows to find the extra structure of a nontrivial principal bundle in our space of solutions. That is, locally, the state space can be viewed as a Cartesian product $\mathbb{S}^2 \times \mathbb{S}^1$. Furthermore, it gives a rigorous argument as to why the states are invariant under change of phase transformations. That is, the structural group of the principal bundle is $U(1)$. This means that the physical states of the qubit are homeomorphic to $\mathbb{S}^2/\mathbb{S}^1$, which is the orbital space. As an additional result, it was obtained that the unitary operators that act over $\mathbb{S}^3$ are a particular case of a linear fractional transformation, a Möbius transformation, when viewed in the extended complex plane $\mathbb{C}_\infty$. Therefore, each unitary operator should translate into combinations of translations, dilations, and inversions.

Hence, the present work is evidence of the profound connection of mathematics and physics, which can be traced even to related fields such as information theory and computation. These insights are not only foundational to quantum mechanics, but also resonate with principles in

quantum information theory, where the structure of qubit state spaces and phase invariance underlies key concepts such as quantum entanglement and quantum error correction.

Although detailed connections may not always lead directly to applications, they provide the fundamental reasons for many physical phenomena and their properties, inspiring further research and exploration. The identification of topological and geometric structures within quantum mechanics offers a fertile ground for future research, potentially leading to breakthroughs in quantum information processing and topological quantum computing. The challenge now lies in fully harnessing these mathematical structures to address practical problems in quantum technology, ultimately contributing to advancing both fundamental science and its applications.

Appendices

A | TOPOLOGICAL CONCEPTS

In this appendix, we will introduce briefly some mathematical concepts needed to better understand the connection between the space of solutions of the qubit system and the sphere. We will begin with some basic definitions and properties from topology that led us to the equivalence of the sphere with antipodal points and the projective plane.

A.1 Topological spaces

Definition A.1.1 (Topological space). *Let X be a set. A topology on X is a collection τ of subsets of X satisfying the following properties:*

- X and $\emptyset$ are elements of τ .
- τ is closed under finite intersections: If $U_1, U_2, \dots, U_n$ are elements of τ , then their intersection $U_1 \cap \dots \cap U_n$ is an element of τ .
- τ is closed under arbitrary unions: If $\{U_\alpha\}_{\alpha \in I}$ is any family (finite or infinite) of elements of τ , then its union $\bigcup_{\alpha \in I} U_\alpha$ is an element of τ .

A pair (X, τ) consisting of a set X and a topology τ in X is called a **topological space**. The elements of X are called **points**, and the sets that make up the topology are called **open subsets** of X or just **open sets**.

This way of defining a mathematical space lets us think of important concepts in real analysis or calculus without measuring a distance. For example, the idea of a neighborhood around a point p in the space. If X is a topological space and $p \in X$, a **neighborhood** of p is just an open subset of X containing p . More generally, if $K \subseteq X$, a **neighborhood of the subset K** is an open subset containing K . (In some references, this definition is more general, saying that a neighborhood is a set such that contains an open subset containing p , but for us it will suffice as stated before).

One pathological feature of general topological spaces is that they first allowed a small number of open sets, so the neighborhoods do not have the same intuitive meaning to which we are used, as in $\mathbb{R}^n$. To solve that, we define a property that ensures this doesn't happen, the Hausdorff property. A topological space X is said to be a **Hausdorff space** if, given any pair of

distinct points $p_1, p_2 \in X$, there exist neighborhoods U_1 of p_1 and U_2 of p_2 with $U_1 \cap U_2 = \emptyset$. This property is often summarized by saying 'points can be separated by open subsets'.

Another concept of great importance in the notion of topological spaces is that of closed subsets. Let us define it as follows: If X is a topological space, a subset $F \subseteq X$ is said to be a **closed subset of X** or **closed set** if its complement $X \setminus F$ is an open subset of X . From the definition, we can derive the following basic properties of closed sets:

- X and $\emptyset$ are closed subsets of X ;
- Any union of finitely many closed subsets of X is a closed subset of X .
- Any intersection of arbitrarily many closed subsets of X is a closed subset of X .

Notice that the properties are almost identical to those that define a topology but with open sets. Moreover, a result that can be proved, and it is useful to prove if a subset $A \subseteq X$ is a closed subset, tells us that A is closed if every point of $X \setminus A$ has a neighborhood contained in $X \setminus A$. One way to inherit a topology from a topological space X is to use the subspace topology.

Definition A.1.2 (Subspace Topology). Let X be a topological space, and let $S \subseteq X$ be any subset. We define the topology τ_S on S by

$$\tau_S = \{U \subseteq S \mid U = S \cap V \text{ for some open subset } V \subseteq X\}.$$

In other words, the open subsets of τ_S are the intersections with S of the open subsets of X . The topology τ_S is called the **subspace topology** on S .

A subset of a topological space X , considered a topological space with the subspace topology, is called a **subspace of X** .

One fundamental function in elementary set theory, called the inclusion map, is closely related to this type of topology. In general, we say that: If $S \subseteq X$ is a subset, there is a function $\iota_S : S \rightarrow X$ called the **inclusion map of S in X** , given by $\iota_S(x) = x$ for $x \in S$. We sometimes use the notation $\iota_S : S \hookrightarrow X$ to emphasize that this is an inclusion map. We will talk about one of its properties in the next section.

A.2 Functions between topological spaces

Now, let us briefly discuss how the mappings relate one topological space with another. For instance, one of the main properties we encounter when we work in metric spaces, in particular, with Euclidean spaces like $\mathbb{R}^n$, is the continuity of a function. This property can be generalized to a topological space with the following definition.

Definition A.2.1 (Continuity). If X and Y are topological spaces, a map $f : X \rightarrow Y$ is said to be continuous if for every open subset $U \subseteq Y$, its preimage $f^{-1}(U)$ is open in X .

We can rewrite the previous definition with the language of closed sets by asking the same questions as they do with their counterparts. Let us now consider the following basic but important properties of continuous maps that will be of importance in the further analysis.

□ **Proposition A.2.1.** Let X, Y and Z be topological spaces:

1. The identity map $Id_X : X \rightarrow X$ is continuous.
2. If $f : X \rightarrow Y$ is continuous, the restriction of f to any open subset of X is also continuous.
3. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then so is their composition $g \circ f : X \rightarrow Z$.

Proof.

1. This proof is straightforward because the identity map is a special bijection from X to X . This means that for every $A \subseteq X$, $f^{-1}(A) = A$. It follows that if A is open, then $f^{-1}(A)$ will be open.
2. Let us suppose that $f : X \rightarrow Y$ is a continuous function and A an open subset of X . So, the restriction of f in A is a map $f|_A : A \rightarrow Y$. As f is continuous, for every open subset $U \subseteq Y$, $f^{-1}(U)$ is an open subset of X . But $f^{-1}|_A(U) = f^{-1}(U) \cap A$. But, by the closure of intersections of open sets, $f^{-1}|_A(U)$ it is an open subset in X . Then $f|_A$ is continuous.
3. Let there be $f : X \rightarrow Y$, $g : Y \rightarrow Z$ continuous functions. Then for every $U \subseteq Z$ open, $g^{-1}(U)$ is open in Y . As f is continuous, $f^{-1}(g^{-1}(U))$ is an open set of X . That is, $(g \circ f)^{-1}(U)$ is an open set of X . It follows that $g \circ f$ is a continuous function. □

The most fundamental relationship we can find between the topological spaces given by continuous maps is that of a homeomorphism. This kind of relationship is of great importance because many topological features or properties of a topological space are preserved.

Definition A.2.2 (Homeomorphism). If X, Y are topological spaces, a **homeomorphism** from X to Y is a bijective map $\varphi : X \rightarrow Y$ such that both φ, φ^{-1} are continuous. If there exists a homeomorphism between X and Y , we say that they are homeomorphic or topologically equivalent. This will be denoted as $X \approx Y$.

An important kind of homeomorphism is the topological embedding, that is, an injective continuous map that is a homeomorphism onto its image (in the subspace topology), is called a **topological embedding** (or just an embedding). If $f : A \rightarrow X$ is such a map, we can think of the image set $f(A)$ as a homeomorphic copy of A inside X .

Other types of functions that are very useful are open and closed maps.

Definition A.2.3 (Open and Closed maps). A map $f : X \rightarrow Y$ (continuous or not) is said to be an **open map** if it takes open subsets of X to open subsets of Y ; in other words, if for every open subset $U \subseteq X$, the image $f(U)$ is open in Y . It is said to be a **closed map** if it takes closed subsets of X to closed subsets of Y .

The last property that will be introduced in this section is about the inclusion map. To keep this section as brief as possible, we will not mention the proof, but you can find it in [22].

□ **Proposition A.2.2.** If S is a subspace of the topological space X , the inclusion map $\iota_S : S \hookrightarrow X$ is continuous.

A.3 Quotient spaces

One powerful way to construct new topological spaces from a specific space that we already know is via the quotient topology. This method provides us with a way to identify some points in a given topological space with each other, to obtain a new, smaller space.

Definition A.3.1 (Quotient topology). Let X be a topological space, Y be any set, and $q : X \rightarrow Y$ be a surjective map. We define a topology on Y by declaring that a subset $U \subseteq Y$ is open if and only if $q^{-1}(U)$ is open in X . This is called the **quotient topology** induced by the map q .

The map q aforementioned is of great importance for some results that we will use later, so let us remark it with the following definition.

Definition A.3.2 (Quotient map). If X and Y are topological spaces, a map $q : X \rightarrow Y$ is called a **quotient map** if it is surjective and Y has the quotient topology induced by q .

Once q is known to be a surjective map, it suffices to see that V is open in Y if and only if $q^{-1}(V)$ is open in X to say that it is a quotient map. It is immediate from this that every quotient map is continuous. The application of the quotient topology that we are interested in is the quotient space. Let $\sim$ be an equivalence relation on a topological space X . For each $p \in X$, let $[p]$ denote the equivalence class of p , and let $X/\sim$ denote the set of equivalence classes. Let $q : X \rightarrow X/\sim$ be the natural projection sending each element of X to its equivalence classes. Then $X/\sim$ together with the quotient topology induced by q is called the **quotient space**. The same concept can be applied to any partition of X .

Example (Projective space): Let us define $\mathbb{P}^n$, the **projective space of dimension n** , to be the set of 1-dimensional linear subspaces (lines through the origin) of a vector

space such as $\mathbb{R}^n$ or $\mathbb{C}^n$, in such cases, we call the **real or complex projective space**, respectively. And we denote $\mathbb{RP}^n$ or $\mathbb{CP}^n$ depending on the situation. There is a natural map

$$q : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$$

defined by sending a point x to its span. We topologize $\mathbb{RP}^n$ by giving it the quotient topology with respect to this map. The 2-dimensional projective space $\mathbb{RP}^2$ is usually called the **projective plane**. Similarly, we topologize the complex case $\mathbb{CP}^n$, but using the quotient $(\mathbb{C}^{n+1} \setminus \{0\} / \mathbb{C}^*)$, where $\mathbb{C}^*$ is the group of non-zero complex numbers acting by scalar multiplication.

The projective space can also be viewed in another way. If we define an equivalence relation on $\mathbb{R}^{n+1} \setminus \{0\}$ by declaring two points x, y equivalent if $x = \lambda y$ for some non-zero real number λ , then there is an obvious identification between $\mathbb{RP}^n$ and the set of equivalence classes. Under this identification, the map q defined above is just the map sending a point to its equivalence class.

We can now introduce other concepts to explore the conditions under a map $q : X \rightarrow Y$ between topological spaces is a quotient map. Suppose $q : X \rightarrow Y$ is a map. Any subset of the form $q^{-1}(y) \subseteq X$ for some $y \in Y$ is called a **fiber** of q . A subset $U \subseteq X$ is said to be **saturated with respect to q** if $U = q^{-1}(V)$ for some subset $V \subseteq Y$. Although quotient maps are not generally open maps, there is an alternative characterization of quotient maps in terms of saturated open or closed subsets.

 **Proposition A.3.1.** *A continuous surjective map $q : X \rightarrow Y$ is a quotient map if and only if it takes saturated open subsets to open subsets, or saturated closed subsets to closed subsets.*

Proof.

- ($\Rightarrow$) Let q be a quotient map, then $U = q^{-1}(V)$ is open if and only if V is open. But q is surjective, so $q(U) = q(q^{-1}(V)) = V$. Consequently, q takes saturated open subsets to open subsets.
- ($\Leftarrow$) Let q be a continuous and surjective map that takes open saturated subsets to open subsets. To see that q is a quotient map, let us see if it satisfies its definition. Let $V \subseteq Y$ be open. Because q is continuous, $q^{-1}(V)$ is also open. Now, suppose that $U = q^{-1}(W) \subseteq X$ is an open subset. Then, U is an open saturated subset. Therefore, $q(U) = q(q^{-1}(W)) = W$ is open. Then q is a quotient map.

□

From this it follows that if $q : X \rightarrow Y$ is a continuous surjective map that is also an open or closed map, then it is a quotient map. This is because an open map takes all open subsets to open subsets, in particular, saturated open subsets.

An important result or property associated with the quotient map is the characteristic property of the quotient topology. This states the following.

■ **Theorem A.3.1 (Characteristic Property of the Quotient Topology).** Suppose X and Y are topological spaces and $q : X \rightarrow Y$ is a quotient map. For any topological space Z , a map $f : Y \rightarrow Z$ is continuous if and only if the composite map $f \circ q$ is continuous:

$$\begin{array}{ccc} X & & \\ q \downarrow & \searrow f \circ q & \\ Y & \xrightarrow{f} & Z \end{array}$$

Proof. Let be $U \subseteq Z$ an open subset. Then $f^{-1}(U)$ is open in Y if and only if $q^{-1}(f^{-1}(U)) = (f \circ q)^{-1}(U)$ is open in X . $\square$

In about the preceding theorem, we say that f **passes to the quotient** or **descends to the quotient**.

The next theorem is one of the most important consequences of the characteristic property of the quotient topology. It tells us how to define continuous maps from a quotient space.

■ **Theorem A.3.2 (Passing to the Quotient).** Suppose $q : X \rightarrow Y$ is a quotient map, Z is a topological space, and $f : X \rightarrow Z$ is any continuous map that is constant on the fibers of q (that is, if $q(x) = q(x')$ then $f(x) = f(x')$). Then there exists a unique continuous map $\tilde{f} : Y \rightarrow Z$ such that $f = \tilde{f} \circ q$.

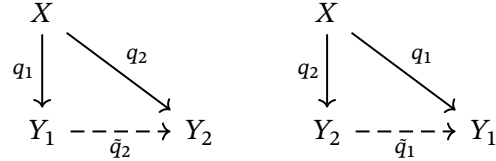
$$\begin{array}{ccc} X & & \\ q \downarrow & \searrow f & \\ Y & \xrightarrow{\tilde{f}} & Z \end{array}$$

Proof. Let be $y \in Y$. Then, because q is surjective, there exists $q(x) = y$, for some $x \in X$. Now, let us define $\tilde{f}$ as $\tilde{f}(y) = f(x)$ for any such x . Let us see if $\tilde{f}$ is well defined. If $x, x' \in q^{-1}(y)$, then $y = q(x) = q(x')$ and that implies, by hypothesis, that $f(x) = f(x') = \tilde{f}$. Let us show that $\tilde{f}$ is unique. Suppose that there exists $g : Y \rightarrow Z$ such that $g \circ q = f$. Then, for some $y = q(x)$, $g(y) = g(q(x)) = f(x) = \tilde{f}(x)$. Therefore, $g = \tilde{f}$. Moreover, from the characteristic property of the quotient topology, it follows that $\tilde{f}$ must be continuous. $\square$

The next consequence of the characteristic property of the quotient topology tells us that quotient spaces are uniquely determined up to homeomorphism by the identifications made by their quotient maps.

■ **Theorem A.3.3 (Uniqueness of Quotient Spaces).** Suppose $q_1 : X \rightarrow Y_1$ and $q_2 : X \rightarrow Y_2$ are quotient maps that make the same identifications (that is, $q_1(x) = q_1(x')$ if and only if $q_2(x) = q_2(x')$). Then, there is a unique homeomorphism $\varphi : Y_1 \rightarrow Y_2$ such that $\varphi \circ q_1 = q_2$.

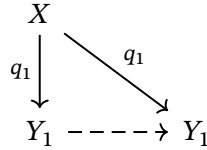
Proof. By the preceding theorem, both q_1 and q_2 pass uniquely to the quotient as in the following diagrams.



Since both diagrams commute, it follows that

$$\tilde{q}_1 \circ (\tilde{q}_2 \circ q_1) = \tilde{q}_1 \circ q_2 = q_1 \quad (\text{A.1})$$

Now, consider the following diagram:



If the dashed arrow is interpreted as $\tilde{q}_1 \circ \tilde{q}_2$, then (1.1) shows that the diagram commutes; it also obviously commutes when the dashed arrow is the identity map of Y_1 . By the uniqueness provided in Theorem 1.3.2, these maps must be equal. Similarly, $\tilde{q}_2 \circ \tilde{q}_1$ is the identity in Y_2 . Thus $\varphi = \tilde{q}_2$ is the required homeomorphism, and its unique because of Theorem 1.3.2. $\square$

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